Automatic Continuity

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Thm (Dudley, 1960s)
G Polish group, $f: G \to \mathbb{Z}$ homomorphism
$\Rightarrow f$ is continuous.

Def. G^{Polish} satisfies automatic continuity (AC) if $\forall$ separable group H and every homo $G \xrightarrow{f} H$ it follows that f is continuous.

Non-ex $\mathbb{R} = \bigoplus \mathbb{Q} \longrightarrow \mathbb{Q}$

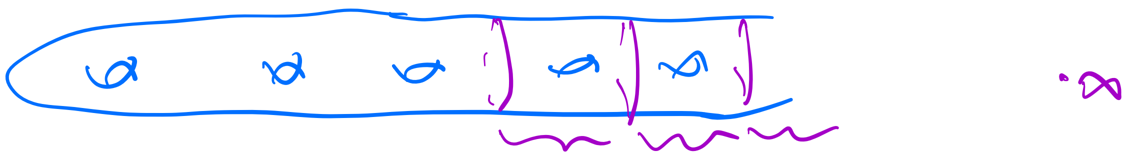
Kallman $GL_n\mathbb{R} \hookrightarrow S_\infty = \mathrm{Homeo}(\cdot \; \cdot \cdot \cdot)$

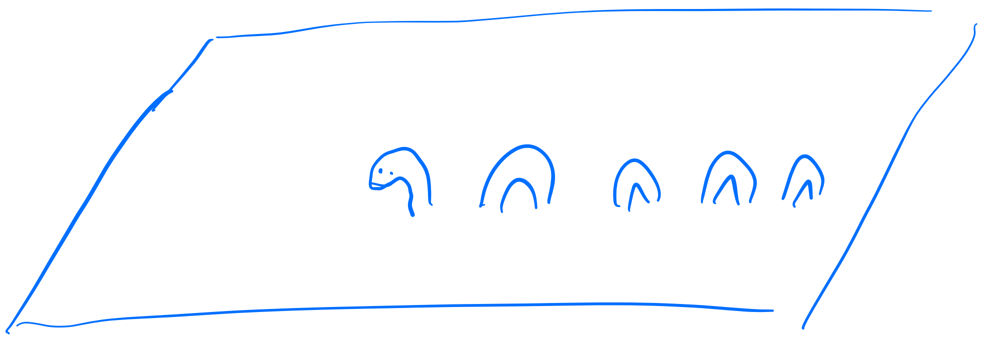
Ex. Rosendal Homeo(compact surfaces)

Rosendal - Solecki Homeo(Cantor) is AC

K. Mann MCG(compact surface − Cantor set)

<u>Non-ex</u> (Domat)





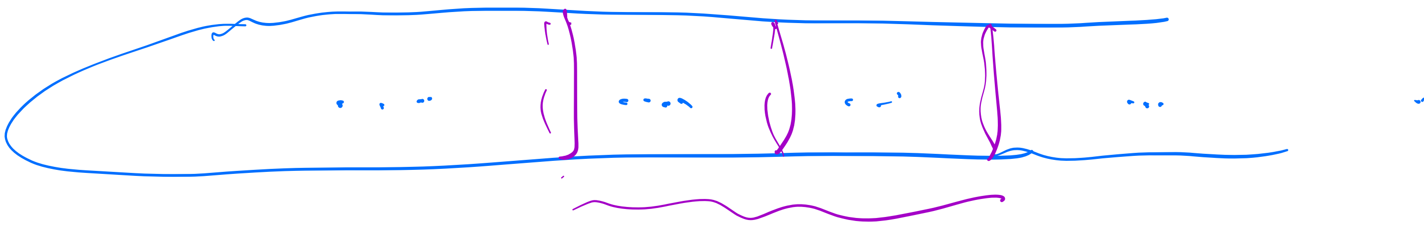
compact metrizable totally disc.

<u>Thm 1</u> X <u>stable</u> Stone space $\Rightarrow$ $\mathrm{Homeo}(X)$ satisfies AC

<u>Thm 2</u> Σ infinite type <u>stable</u> surface.

$\mathrm{Homeo}(\Sigma)$, $\mathrm{MCG}(\Sigma)$ satisfy AC

$\Leftrightarrow$ every end of Σ is telescoping.

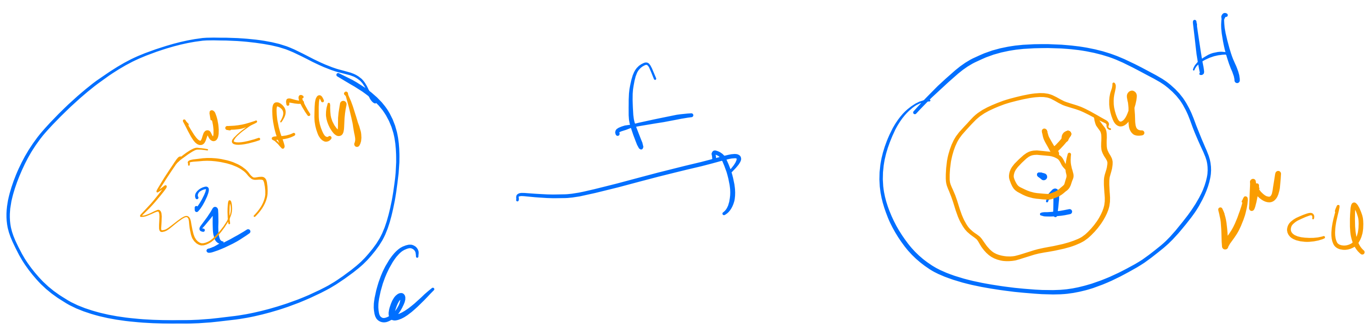


Def. G is <u>Steinhaus</u> if $\exists\, N$ s.t.

$\forall$ symmetric subset $W \subset G$ s.t. $G = \bigcup_{i=1}^{\infty} g_i W$ some $g_i \in G$

it follows that W^N contains a nbhd of 1.

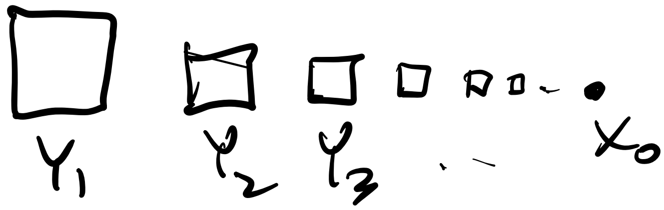
Prop Steinhaus $\Rightarrow$ AC



$W^N \supseteq$ nbhd of 1, $f(W^N) \subseteq U$.

<u>Def of stable</u> X Stone space, $x_0 \in X$.

A stable nbhd of x_0 is a clopen set $\Omega \ni x_0$ and its



$Y_n \xrightarrow[\text{as a clopen}]{} Y_{n+1}$

X is stable if every pt has a stable nbhd.

<u>Ex.</u>

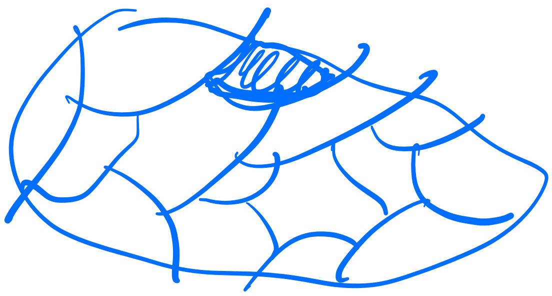
$Y_i = \text{pt}$

Cantor set $Y_i = \text{Cantor set}$

<u>Ex.</u> G Steinhaus, W as in the def.

$\Rightarrow$ $\overline{W}^2$ contains a nbhd of 1.

Local Thm Suppose X is a stable nbhd of some $x_0 \in X$ and W^2 is dense in $\mathrm{Homeo}(X)$



Goal: $\exists N$ s.t. $W^N = \mathrm{Homeo}(X)$

① Fragmentation

$$X = \coprod_{i=1}^{\infty} Y_i \sqcup \{x_0\}$$

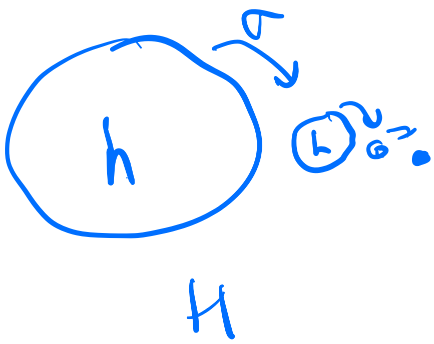
Brick:

$$B = \coprod_{k=1}^{\infty} Y_{i_k} \quad \text{s.t. } \{i_k\},\ \mathbb{N} - \{i_k\} \text{ both infinite}$$

Prop Every homeo of X can be written as the composition of 4 homeos each supported on a brick.

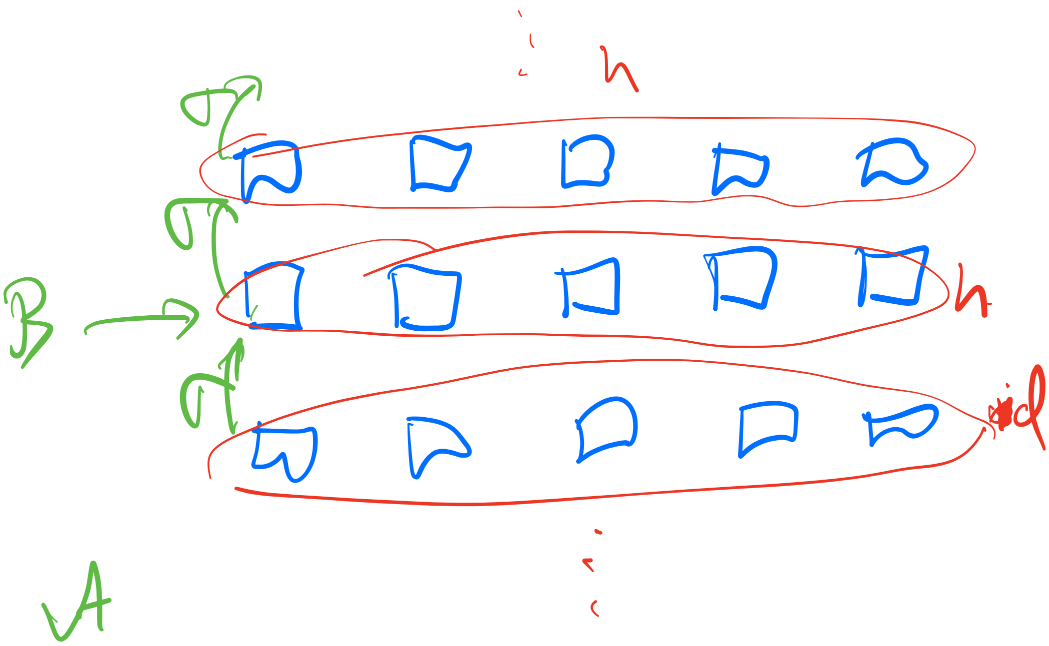
② Finding commutators

Mather: every compactly supported $h \in \mathrm{Homeo}(\mathbb{R}^2)$ is a commutator.



$[H, \sigma] = h$

Use a bijection $\mathbb{N} \Longleftrightarrow \mathbb{Z}^2$



<u>Prop</u> $B < A$ bricks ($A - B$ also a brick). Then every homeo supported on B is the commutator of homeos supported on A.

③ Diagonalization

$A^{\text{brick}} > A_1, A_2, \ldots$ pairwise disj. subbricks

Then $\exists\, i$ st any homeo in $G(A_i)$ extends to a homeo supported on A that belongs to $g_i W$.

supported on A_i

Moreover, it extends to a homeo supported on A that belongs to W^2.

Pf. If not, let h_i be a counterexample

- $\frac{h_1}{A_1} \quad \frac{h_2}{A_2} \quad \cdots\cdots \quad \in g_i W$ some i

- $h = \tilde{i}h. \quad = \tilde{i}\tilde{h}$

$$g_iW \quad g_iW \subseteq W^2$$

Cor

$$\underset{\tilde{v} \in W^2}{A} > \underset{\tilde{u} \in W^2}{B} \supset \underset{u,v}{\mathcal{C}} > \underset{h}{\mathcal{Z}}$$

$h = [u,v] = [\tilde{u}, \tilde{v}]$

$h \in G(Z) \implies h \in W^8$

④ Pigeonhole.

Every homeo supported on any brick is in W^{20}.

A brick, $h \in G(A)$

$Z \subset X - A$ s.t. $G(Z) \subset W^8$.

$Z = \coprod Z_i$ boxes

Fix $\{ \Lambda_\alpha \subset \mathbb{N} \}_\alpha$ uncountably many infinite subsets s.t. $\Lambda_\alpha \cap \Lambda_\beta$ finite $\alpha \neq \beta$.

$Z_\alpha = \coprod_{i \in \Lambda_\alpha} Z_i$

f_α homeo that interchanges Z_α and its complement

$\exists\ \alpha, \beta \quad f_\alpha, f_\beta \in g_i W$

$f_\beta^{-1} f_\alpha \in W^2$

$$F = \overbrace{(f_\beta^{-1} f_\alpha)}^{W^2} h \overbrace{(f_\beta^{-1} f_\alpha)}^{W}{}^{-1} \text{ is}$$

supported in $\underline{f_\beta^{-1} f_\alpha(A)}_{\subseteq Z_\alpha} \subseteq Z_\beta \cup$ finitely many Z_i

$\subseteq Z$

$\therefore F \in W^8 \quad \therefore h \in W^{12}$